MATHEMATICS - QUESTION BANK

SSC 100 DAYS ACTION PLAN

1. REAL NUMBERS

[9 Marks] [1 + 8 = 9 M]

LEVEL-1 : RISING STAR

8 MARK QUESTIONS

1. Is $\sqrt{5}$ Irrational ? Justify your Answer.
2. Is $\sqrt{2}$ Irrational ? Justify your Answer. (Also make them to practice for $\sqrt{3}, \sqrt{7}$ also).

LEVEL-2 : SHINING STAR

8 MARK QUESTIONS

1. Examine whether $3+2\sqrt{5}$ is Irrational or not.
2. Is $5-\sqrt{3}$ is Irrational ? Justify your Answer.
3. Examine whether $6+\sqrt{2}$ is Irratonal or not.
4. Give an example for an irrational number. Also prove that it is an Irrational.

1 MARK QUESTIONS

1. Explain the fundamental theorem of arithmetic.
2. HCF (20, 35) = 5 then LCM (20, 35) = ... ( )

   A) 140 B) 75 C) 250 D) 350
3. LCM (a, b) = a × b (a ≠ b) then HCF (a, b) = ... ( )

   A) ab B) a C) b D) 1
4. HCF (a, 20) = 2, LCM (a, 20) = 60 then a = ... ( )

   A) 2 B) 6 C) 12 D) 24
5. Unit digit in the Product of 6^n, (n is natural number) ( )

   A) 9 B) 3 C) 6 D) 2
6. A, B are positive integers $A = p^3q^2$, $B = pq^3$ (p, q are prime numbers) then HCF (A, B) = ...

   A) pq^2 B) p^2q^2 C) p^3q^3 D) 1
7. If n is any natural number, then which the following expression ends with zero ( )

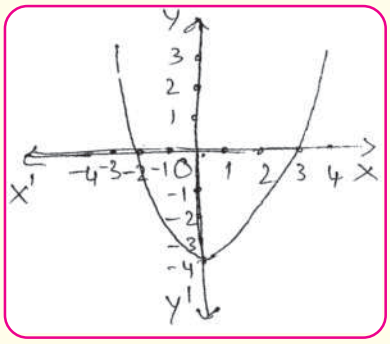
   A) $(3 \times 2)^n$ B) $(4 \times 3)^n$ C) $(2 \times 5)^n$ D) $(6 \times 2)^n$
9. Express 153 as product of prime factors .....
9. Classify $\sqrt{3}, \sqrt{4}, \pi, 2.43$ in to rational and Irrational numbers.
10. If P is a prime number P divides a^2 then P also dvides .....

2. POLYNOMIALS

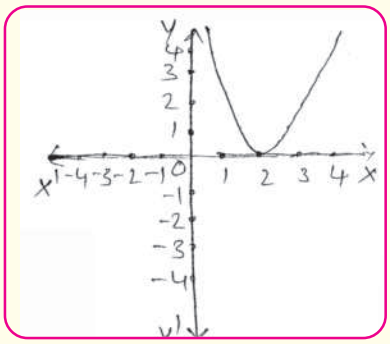
[8 Marks] [1 + 1 + 2 + 4 = 8 M]

LEVEL-1 : RISING STAR

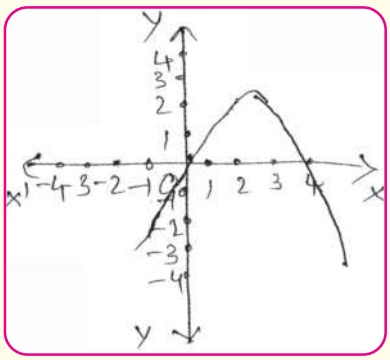
4 MARKS QUESTIONS



1. By observing the graph answer the following questions.
   a. What is the shape of graph in the figure ?
   b. How many zeroes are there for that polynomial ?
   c. Write the zeroes of the polynomial.
   d. Find the product of zeroes of the polynomial.



2. By observing the graph answer the following questions.
   a. What is the name of the polynomial in the graph ?
   b. How many zeroes it has
   c. Write the zeroes of polynomial
   d. Find the sum of zeroes of polynomial.



3.. By observing the graph answer the following questions.
   a. What is the shape of graph in the figure ?
   b. Write the zeroes of the polynomial.
   c. Find the sum of zeroes of the polynomial.
   d. Find the product of zeroes of polynomial.

2 MARKS QUESTIONS

1. Generate a quadratic polynomial whose sum and product of zeroes are $\frac{1}{4}, 1$
2. Create a quadratic polynomial whose sum and product of a zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

LEVEL-2 : SHINING STAR

4 MARKS QUESTIONS

1. Find the zeroes of the polynomial $x^2 - 3$ and verify the relationship between zeroes and coefficients.

2 MARKS QUESTIONS

1. Find the zeroes of quadratic polynomial $P(x) = x^2 - 2x - 8$
2. Find the sum and product of zeroes of the polynomial $P(x) = 4x^2 + 8x$
3. Generate a quadratic polynomial whose zeroes are – 3, 4.

1 MARK QUESTIONS

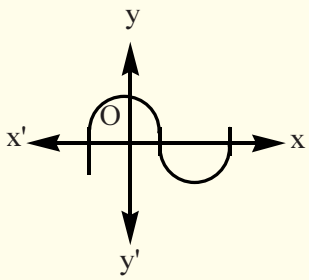
1. The graph of the polynomial $P(x) = 3x - 1$ meets x-axis at the point .....
2. Assertion (A) : The polynomial $P(x) = 4x + 3$ is a linear polynomial ( )

   Reason (R) : The general form of linear polynomial is $ax + b$

   Which of the followiong is true.

   A) Both A, R are true B) Both A, R are false

   C) A is true, R is false D) A is false, R is true
3. The degree of the polynomial of $P(x) = 5x^3 - 4x^2 + x - \sqrt{2}$ is .....
4. Draw the graph of $P(x) = 2x + 3$.
5. The zeroes of $x^2 - 4$ are 2 and – 2 (True/False)
6. The coefficient of x^2 in $P(x) = 7x^3 - 6x^2 + 5x + 8$ is .....
7. The graph of polynomial is shown in the figure then the number of zeroes are .....



8. Product of zeroes of $2x^2 + 6x + m$ is – 1 then m = ...
9. Draw the rough diagram of quadratic polynomial.
10. Write a polynomial whose zeroes are 2 and 4.

3. PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

[9 Marks] [1 + 8 = 9 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Solve the following pair of linear equation graphically

   $x - y = 4; x - 2y = 6.$

2. Solve the following pair of linear equation graphically

   $2x + 3y - 6 = 0 ; x + y + 3 = 0$

3. Solve the following pair of linear equation graphically

   $2x + y = 6 ; 4x - 2y = 4$

4. Form the pair of linear equations in the following situation and find their solution graphically.

   5 pencils and 7 pens together cost 50 Rupees, where as 7 pencils and 5 pens together cost 46 Rupees.

5. Solve the following pair of linear equation graphically

   $x + 3y = 6 ; 2x - 3y = 12$

6. "10 students of class X took part in a Mathematics Quiz. If the number of girls is 4 more than the more of boys". Then form the pair of linear equations in the above situation and find the number of boys and girls took part in the quiz by graphically.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Form the pair of linear equations for the following problem and find the solution by graphical method "sum of two numbers is 10 and their difference is 2".

2. Half the perimeter of a rectangular garden , whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden graphically.

3. Solve the following pair of linear equations graphically

   $2x + y - 5 = 0 , 3x - 2y - 4 = 0$

4. Draw the graphs of the equation $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the X-axis and shade the triangular region.

1 MARK QUESTIONS

1. Write the general form of linear equation in two variables.
2. "The cost of 2 pens and 5 pencils is Rs. 20". Express this data as a linear equation.
3. **Match the following.**

   $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are pair of linear equations.

   i) Unique solution ( ) a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

   ii) No solution ( ) b) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

   iii) Infinity many solutions ( ) c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

   A) 1-a, 2-b, 3-c B) i-b, ii-c, iii-a
   B) 1-c, ii-b, iii-a D) i-b, ii-a, iii-c
4. Write another linear equation is parallel to $3x - 2y + 4 = 0$
5. On comparing ratios $\frac{a_1}{a_2}, \frac{b_1}{b_2}, \frac{c_1}{c_2}$ find out whether the following pair of linear equations are consistent or inconsistent.

   $2x - 3y = 8$; $4x - 6y = 9$
6. Find the value of k for which the following equations have infinitely many solutions

   $4x + 5y = 3$

   $kx + 15y = 9$
7. Draw the rough diagrams of pair of linear equations in two variables which have unique solution and infinitely many solutions.
8. Form the pair of linear equations for the following data "The larger of two supplementary angles exceeds the smaller by 18°.
9. **Assertion (A) :** The pair of equations $2x - 3y = 1$ and $3x + y = 7$ are intersecting lines.

   Reason (R) : The pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are intersecting if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. Now, choose the correct answer.

   A) Both A and R are true. B)Both A and R are false.

   C) A is true, but R is false. D)A is false, but R is true.
10. **Assertion 1 (A_1) :** Pair of linear equations have infinitely many solutions if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

    Assertion 2 (A_2) : Pair of linear equations have no solutions if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

    Now choose the correct answer.

    A) Both A_1 and A_2 are true. B) Both A_1 and A_2 are false.
    C) A_1 is true, A_2 is false. D) A_1 is false, A_2 is true.

4. QUADRATIC EQUATIONS

[7 Marks] [1 + 2 + 4 = 7 M]

LEVEL-1 : RISING STAR

4 MARKS QUESTIONS

1. State the conditions under which a quadratic equation will have

   i) Two real and distinct roots ii) Two equal roots

2. Find the discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ and hence find the nature of roots. Find them if they are real.

3. State the nature of roots of the following quadratic equations.

   i) $2x^2 - 3x + 5 = 0$ ii) $2x^2 - 6x + 3 = 0$

4. Express the roots of the quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$ by quadratic formula method.

5. Find the nature of roots of quadratic equation $2x^2 + x - 6 = 0$. If the real roots exist find them.

6. Find the value of 'K' for the quadratic equation $Kx(x - 2) + 6 = 0$ if it has two equal roots.

2 MARKS QUESTIONS

1. Develop the following situation in the form of quadratic equation. "The product of two consecutive positive integers is 306".

2. Analyze whether $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation or not ?

3. Detect the nature of roots of $2x^2 + 3x + 5 = 0$ by calculating discriminant.

4. Find the value of 'K' for quadratic equation $2x^2 + Kx + 3 = 0$ if it has two equal real roots.

5. Find the nature of roots of quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$

6. Analyse whether $x^2 + 3x + 1 = (x - 2)^2$ is a quadratic equation or not ?

7. Detect the roots of quadratic equation $6x^2 - x - 2 = 0$

LEVEL-2 : SHINING STAR

4 MARKS QUESTIONS

1. Find the numbers whose sum is 27 and product is 182.

2. Find two consecutive positive integers, whose sum of squares is 365.

3. Is it possible to design a rectangular mango grove whose length is twice its breadth and the area is 800 m^2. If so find its length and breadth.

1 MARK QUESTIONS

1. Write the standard form of a quadratic equation.
2. Form a quadratic equation whose roots are real and equal.
3. Form a quadratic equation which has 2 as one of its roots.
4. Create the following situation as quadratic equation" – The product of two consecutive Integers is 15".
5. Generate the following situation as quadratic equation "The sum of number and its reciprocal is $\frac{1}{2}$.
6. Form a quadratic equation whose roots are reciprocal to each other.
7. Create a quadratic equation which has 1 as one of its roots.
8. Create a quadratic equation which has 2 and 3 as roots.
9. Generate a quadratic equation which has equal roots.
10. Write the quadratic formula for finding roots of quadratic equation.

5. ARITHMATIC PROGRESSIONS

[13 Marks] [1 + 4 + 8 = 13 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. If the sum of the first 'n' terms of an AP is $4n - n^2$. What is the first term, what is the sum of first two terms ? What is the second term ? Similarly find 3rd, 10th and the nth terms.
2. The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP.
3. Subba Rao started work in 1995 at an annual salary of ₹5000 and received an increment of ₹200 each year. In which year did his income reach ₹ 7000 ?
4. If the sum of first 7 terms of an AP is 49 and that of 17 terms is 289, find the sum of first 'n' terms.
5. A sum of ₹700 is to be used to give seven cash prizes to students of a school for their over all academic performance. If each prize is ₹20 less than its preceding prize, find the value of the each of the prizes.

4 MARKS QUESTIONS

1. In a flower bed, there are 23 rose plants in the first row, 21 in the second, 19 in the third and so on. There are 5 rose plants in the last row. How many rows are there in the flowerbed?
2. What is the nth term of AP ? Check – 150 is a term of the A.P : 11, 8, 5, 2, ...
3. Write the formulas for the following AP : $a_1, a_2, a_3, \ldots\ldots a_n$
   A) n^{th} term a_n
   B) Sum of first n terms (S_n)
   C) Common difference
   D) If first term is a, last term is a_n sum of first n terms (S_n)
4. How many terms of AP 24, 21, 18, ... must be taken so that their sum is 78 ?
5. How many multiples of 4 lie between 10 and 250 ?
6. What is the formula for sum of first n terms if first and last terms are given. Find the sum of first 40 positive integers divisible by 6.
7. What is the nth term of A.P. Find the 20th term from the end of A.P. 3, 8, 13, ... 253

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. A manufacturer of TV sets produced 600 sets in the third year and 700 sets in the seventh year. Assuming that the production increases uniformly by a fixed number every year.
   Find (i) The production in the first year
   (ii) Find the production in the tenth year.
   (iii) Total production in first 7 years.

2. In a school, students thought of planting trees in and around the schol to reduce air pollution. It was decided that the number of trees, that each section of each class will plant, will be same as the class, in which they are studying e.g. a section of class I will plant one tree, a section of class II will plant 2 trees and soon till class XII. There are three sections of each class. How many trees willbe planted by the students ?
3. 200 logs are stacked in the following manner 20 logs in the bottom row, 19 in the next row, 18 in the next to it and so on. In how many rows are the 200 logs placed and how many logs are in the top row ?

4 MARKS QUESTIONS

1. If the sum of the first 14 terms of an AP is 1050 and its first term is 10, then find the 20th term.
2. What is the formula for sum of first n terms ? Show that a_1, a_2, a_3, ... a_n form an AP. Where an is defined as $a_n = 3 + 4n$. Find the sum of first 15 terms.
3. What is the formula for sum of first n terms ? Find the sum of the odd numbers between 0 and 50.
4. Given a = 3, n = 8, S = 192 find d.
5. How many three digit numbers are divisible by 7 ?
6. Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

1 MARK QUESTIONS

1. What is the common difference of $\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{-3}{2}$?
2. What is the next term of the series $3, 3+\sqrt{2}, 3+2\sqrt{2}, ...$
3. What is the common difference of $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, ...$
4. Fin the value of x if 13, 3x + 4, 13 are in A.P.
5. Find the sum of first 100 natural numbers.
6. What is the common difference of the AP : x – y, x, x + y ..... ?
7. What is the sum of first 'n' even natural numbers.
8. What is the sum of first 'n' odd natural numbers.
9. Find the sum of first 25 odd natural numbers.
10. Find sum of first 10 natural numbers is .....
11. The common difference of AP 3, 1, – 1, – 3, ... is .....
12. The sum of first 'n' natural numbers is .....
13. If n^{th} term of an AP is 3 + 4n then its common difference is .....
14. If a, b, c are in AP then b = ...

15. Find the common difference of $\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}, -\frac{5}{2}$.....

16. Determine the common difference of $\sqrt{27}, \sqrt{108}, \sqrt{243}$ .....

17. Find the sum of first 100 natural numbers.

18. If $a_n = \frac{n}{n+1}$ then find a_{2025}

19. Find the sum of first 1000 positive integers.

20. The 30th term of AP 10, 7, 4, ...

21. Match the following. ( )

p) Sum of first 10 natural numbers	i) 129
q) Sum of first 10 even numbers	ii) 55
r) Sum of first 10 prime numbers	iii) 110

A) p-i, q-i, r-iii B) p-ii, q-i, r-iii

C) p-ii, q-iii, r-i D) p-iii, q-i, r-ii

22. Assertion (A) : 5 is the common difference of AP 11, 16,21 101

Reason (R) : The common difference of an AP $d = a_n - a_{n-1}$

Now choose the correct answer. ( )

A) A is true, R is false B) A is false, R is true

C) Both A and R are true and R is correct explanatin of A

D) Both A and R are true But R is not correct explanatin of A

23. Statement I : 2, 4, 6, 8 ..... is in AP

Statement II : 10, 10, 10, ... is not in AP

A) Both statements are true ( )

B) Statement I is true But Statement II is false

C) Statement I is false But Statement II is true

D) Both Statements are false.

24. If 7 times the 7th term of an AP is equal to 11 times its 11th term then its 18th term.( )

A) 7 B) 11 C) 18 D) 0

25. Which of the following is not form an AP ?

A) 2x, 4x, 6x, 8x, ... B) 5, 0, – 5, – 10, ...

C) 25, 20, 17, ... D) –3, – 6, – 9, ...

6. TRIANGLES

[11 Marks] [1 + 2 + 8 = 11 M]

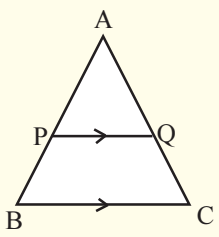
LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

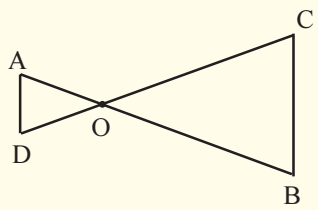
1. State and prove Basic Proportionality Theorem.
2. ABCD is a trapezium in which AB// DC and its daigonals intersect each other at the point O. Show that $\frac{AO}{CO} = \frac{BO}{DO}$
3. If AD and PM are medians of triangles ABC and PQR, respectively where ΔABC is similar to ΔPQR. Prove that $\frac{AB}{PQ} = \frac{AD}{PM}$
4. Prove that if a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.
5. ACBD is a trapezium with AB//DC. Eand F are points on non-parallel sides AD and BC respectively. Such that EF is parallel to AB. Show that $\frac{AE}{ED} = \frac{BF}{FC}$

2 MARKS QUESTIONS

1. Give an example for (i) Similar figures (ii) Non-similar figures.
2. What are the conditions for the similarity of triangles ?
3. State S.A.S criteria in similarity of traingles.
4. State A.A. criteria in similarity of traingles.
5. State S.S.S criteria in similarity of traingles.
6. In given figure, PQ//BC, PQ = 3 cm, BC = 9 cm and AC = 7.5 cm. Find the length of AQ?



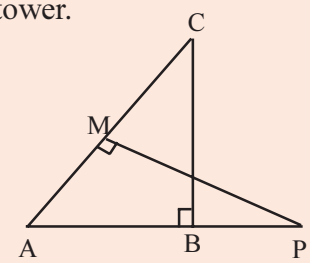
7. In given figure, OA.OB = OC.OD.
   Show that ∠A = ∠C and ∠B = ∠D



LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. A girl of height 90 cm is walking away from the base of a lampo-post at a speed of 1.2 m/sec. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.
2. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.
3. In given figure,e ABC and AMP are two right triangles, right angled at B and M respectively. Prove that
   i) $\Delta ABC \sim \Delta AMP$
   ii) $\frac{CA}{PA} = \frac{BC}{MP}$



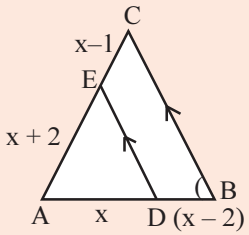
4. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.
5. CM and RN are respectively the medians of ΔABC and ΔPQR. If $\Delta ABC \sim \Delta PQR$.

   Prove that i) $\Delta AMC \sim \Delta PNR$ ii) $\frac{CM}{RN} = \frac{AB}{PQ}$ iii) $\Delta CMB \sim \Delta RNQ$

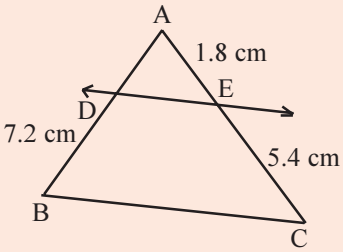
2 MARKS QUESTIONS

1. In the figure, DE//BC, if AD = x, DB = x – 2, AE = x + 2 and EC = x – 1.

   Find the value of x.

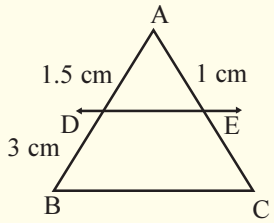


2. In the figure, DE//BC. Find AD.



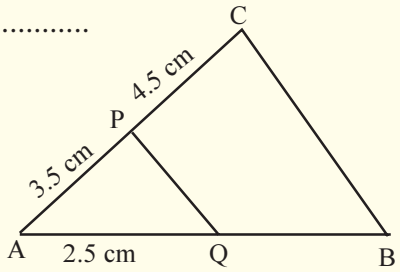
1 MARK QUESTIONS

1. In ΔABC, DE//BC then length of EC = ( )



A) 2 cm B) 2.25 cm C) 3.5 cm D) 1.4 cm

2. In the given figure, PQ//CB, then QB = ... ( )



A) 2.4 cm B) 2.7 cm

C) 3.2 cm D) 4.2 cm

3. Write the statement of Basic proportional theorem. (Thale's Theorem)

4. Statement I : All squares are congruent to each other. ( )
Statement II : All circles are similar
A) Statement I is true, Statement II is false.
B) Statement I is false, Statement II is true
C) Both Statements are true.
D) Both Statements are false.

5. **Statement I :** All similar triangles are congruent. ( )
Statement II : All right angled isosceles triangles are similar.
A) Both Statements are true.
B) Statement I is true and Statement II is false.
C) Statement I is false and Statement II is true
D) Both Statements are false.

6. In ΔABC and ΔPQR, if $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$ then which of the following is true ? ( )

A) ΔCBA ~ ΔPQR B) ΔPQR ~ ΔABC

C) ΔBCA ~ ΔPQR D) ΔPQR ~ ΔCAB

7. Which of the following is not a similarity criterian for two triangles ? ( )

A) S.A.S B) A.A.A

C) A.S.A D) S.S.S

8. If ΔABC ~ ΔXYZ then which of the following is true ? ( )

A) ∠A = ∠Z B) $\frac{AC}{BC} = \frac{XZ}{YZ}$ C) ∠B = ∠Z D) BC = YZ

9. Which of the following statement is true ? ( )
   A) All equilateral triangles are similar B) All circles are similar
   C) All squares are similar D) All the above
10. $\Delta ABC \sim \Delta PQR$ and $\angle A = 70°$ then $\angle Q + \angle R =$ ..... ( )
   A) 180° B) 120° C) 110° D) 70°
11. ΔABC and ΔDEF are similar. $\angle A = 40°$ and $\angle F = 70°$,
   then the value of $\angle B + \angle E$ is ( )
   A) 110° B) 140° C) 100° D) 70°
12. ΔABC and ΔPQR are similar. If AB = 4 cm, PQ = 6 cm, QR = 9 cm and PR = 12 cm then the perimeter of ΔABC is ..... cm. ( )
   A) 15 cm B) 20 cm C) 18 cm D) 24 cm
13. **Statement I :** Two equilateral triangles are always similar.
   Statement II : $\Delta ABC \sim \Delta XYZ$ then AB : XY = AC : XZ ( )
   A) Both Statements are true.
   B) Statement I is true and Statement II is false.
   C) Statement I is false and Statement II is true
   D) Both Statements are false.
14. Two triangles with equal corresponding sides always similar. Is it true or false ?
15. Name the two figures that are always similar.
16. In the triangles ABC and DEF, $\frac{AB}{DE} = \frac{BC}{FD}$ then they will be similar when ( )
   A) $\angle B = \angle E$ B) $\angle A = \angle F$ C) $\angle A = \angle D$ D) $\angle B = \angle D$
17. **Assertion (A) :** D and E are the points of sides AB and AC of ΔABC respectively and DE//BC then AD : DB = AE : EC
   Reason (R) : If a line is drawn parallel to one side of a triangle, it will divide the other two sides in the same ratio. ( )
   A) Both Assertion and Reason are true and reason is the correct explanation of Assertion.
   B) Both Assertion and Reason are true but reason is not the correct explanation of Assertion.
   C) Assertion is true but Reason is false.
   D) Assertion is false but Reason is true.
18. $\Delta ABC \sim \Delta PQR$, $\angle A = 50°$, $\angle Q = 40°$ then ΔQPR is ( )
   A) Isosceles triangle B) Right angled triangle
   C) Equilateral triangle D) Isosceles Right angled trianle
19. **Statement A :** All circles are similar. ( )
   Statement B : All triangles are similar.
   A) Both A and B are true B) Both A and B are false
   C) A is false B true D) A true B false
20. Write the proportionality relation if MN//BC in ΔABC

7. COORDINATE GEOMETRY

[10 Marks] [2 + 8 = 10 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Two students claim to have found the points of trisection of the line segment joining A (2, – 2) and B(– 7, 4) as follows :
   Bharath : (–1, 0) and (– 4,2).
   Vinod : (1, 0) and (4,–2).Who is correct? Justify?
2. Do the points (3, 2), (–2, –3) and (2, 3) form a triangle? If so, name the type of triangle formed:
3. Two students Ramu and Somu solved the problem of finding trisection points of the line segment joining A(4, –1) and B(–2, –3). Ramu got the points as $(2, -\frac{5}{3})$ and $(0, -\frac{7}{3})$ while somu got (2, –2) and (0, –3) Evaluate and decide which one is correct.
4. Check whether the points (4, 5), (7, 6), (4, 3) and (1, 2) taken in order form a parallelogram?
5. Verify whether the points (3, 0), (4, 5), (–1, 4) and (–2,–1) are vertices of rhombus? If so find the area of rhombus

2 MARKS QUESTIONS

1. Write the formula to find the distance between:
   (a) the two points (x_1, y_1) and (x_2, y_2) (b) The origin and the point (x, y)
2. Find the coordinates of mid-point on the line segment joining the points (–2,–2) and (2,–4).
3. Find the distance between the points (2, 3) and (4, 1)
4. Find the coordinates of the point A, where AB is the diameter of a circle whose center is (2, –3) and B is (1, 4).
5. How can you identify the three points A, B and C are collinear?
6. Write the formulae to find:
   (i) The point which divides the joining of the two points (x_1, y_1) and (x_2, y_2) in the ratio $m_1 : m_2$ internally.
   (ii) The mid-point on the line joining the points (x_1, y_1) and (x_2, y_2)

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Q(0, 1) is equidistant from P(5, –3) and R(x, 6) Find the value of 'x'. Determine the lengths of QR and PR
2. Find the coordinates of the points which divides the line segment joining A (–2, 2) and B(2, 8) into four equal parts.

3. In what ratio does the point (–4, 6) divides the line segment joining the points A(–6, 10) and B(3, –8).

2 MARKS QUESTIONS

1. Find the coordinates of mid-point on the line segment joining the points (Cos 0°, 0) and (0, Sin 90°).
2. Find the ratio in which the line segment joining A(1, – 5) and B (–4, 5) is divided by the X-axis?
3. Find the value of 'y 'for which the distance between the points P (2, – 3) and Q(10, y) is 10 units?
4. Find the coordinates of the point which divides the line joining (– 1, 7) and (4, –3) in the Ratio 2 : 3
5. Find the point on X-axis which is equidistant from A (2,–5) and B (–2, 9)?
6. Find the value of 'p', if the points A(6, 1), B(8, 2) C(9,4) and D(p, 3) are the vertices of a parallelogram.
7. What are the coordinates of the point which lies on both the axes?
8. Find the distance between the points (3,0) and (0,4).
9. Write the section formula in internal division for the points (x_1, y_1) and (x_2, y_2) the ratio $m_1 : m_2$
10. What is the distance between the points (a, b) and (a, – b)?
11. The midpoint of on the line joining the points (– 4, 2) and (4,–2) is (0, 2). (True/False)
12. In which ratio that y-axis divide the line joining the points (5, 2) and (3, –2)
13. If the points P and Q are the trisecting points on the line segment AB, what ratio does P and Q divides the line AB?
14. Find the distance between the points (a Cos θ, 0) and (0, a Sin θ)

8. INTRODUCTION TO TRIGONOMETRY

[7 Marks] [1 + 2 + 4 = 7 M]

LEVEL-1 : RISING STAR

4 MARKS QUESTIONS

1. State any three trigonometric Identities.
2. Find the value of

   a) $\sin 60° \cos 30° + \sin 30° \cos 60°$

   b) $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$
3. Write other trigonometric ratios of $\angle A$ interms of sec A
4. Reproduce $\sqrt{\frac{1+\sin A}{1-\sin A}}$ as sec A + tan A (or) Show that $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$.
5. Reproduce $(\sin A + \text{cosec } A)^2 + (\cos A + \sec A)^2$ as $7 + \tan^2 A + \cot^2 A$

2 MARKS QUESTIONS

1. If $\sec \theta = \frac{13}{12}$ determine all trigonometric ratios.
2. If $\sin A = \frac{3}{4}$ determine cos A, tan A
3. If 15 cot A = 8 determine sin A, sec A
4. If $\cot \theta = \frac{7}{8}$ determine $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$
5. If $\sin(A-B) = \frac{1}{2}$, $\cos(A+B) = \frac{1}{2}$ measure the values of A, B
6. If $\tan(A+B) = \sqrt{3}$, $\tan(A-B) = \frac{1}{\sqrt{3}}$ measure the values of A, B

LEVEL-2 : SHINING STAR

4 MARKS QUESTIONS

1. Find the values of

   a) $\frac{\sin 90° \tan 45° + \cos 0° \sec 60°}{\sec^2 46 + \text{cosec}^2 45}$

   c) $\sqrt{2}\sin 45° + \sqrt{3}\cos 30° + \text{cosec } 90°$
2. Write cos A, tan A, sec A interms of sin A

3. Reproduce $(\text{cosec}\ \theta - \cot\theta)^2$ as $\frac{1-\cos\theta}{1+\cos\theta}$ (or) Show that $(\text{cosec}\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$.

4. Reproduce $\frac{\cot A - \cos A}{\cot A + \cos A}$ as $\frac{\text{cosec}\ A - 1}{\text{cosec}\ A + 1}$

2 MARKS QUESTIONS

1. $\sec A\,(1 - \sin A)\,(\sec A + \tan A) = 1$ Justify ?
2. In a right angled triangle ΔABC, $\angle B = 90°$, AB = 5 cm, $\angle ACB = 30°$. Determine the lengths of BC and AC.
3. In a ΔABC, $\angle B = 90°$, If $\tan A = \frac{1}{\sqrt{3}}$ find the value of $\sin A \cos C + \cos A.\ \sin C$

1 MARK QUESTIONS

1. Which trigonometric ratio equal to Adj side/Hypotenuse ? ( )
   A) $\sin\theta$ B) $\cos\theta$ C) $\tan\theta$ D) $\cot\theta$
2. If a right angled triangle. If $\tan A = 1$, then $\angle A =$ ( )
   A) 0° B) 30° C) 45° D) 60°
3. If $\sin\theta = \cos\theta$ then $\tan\theta =$ ( )
   A) 0 B) $\frac{1}{\sqrt{3}}$ C) $\sqrt{3}$ D) 1
4. If $\tan A = 1$, then $2\sin A \cos A =$ ( )
   A) 0 B) $\frac{1}{2}$ C) 1 D) 2
5. $9\sec^2 A - 9\tan^2 A =$ ..... ( )
   A) 0 B) 1 C) 8 D) 9
6. $\text{cosec}\ \theta - \cot\theta = x$ then $\text{cosec}\ \theta + \cot\theta =$ ( )
   A) $-x$ B) $x + 1$ C) $\frac{1}{x}$ D) $-\frac{1}{x}$
7. If 'θ' is an acute angle, As value of 'θ' Increase then value of $\cos\theta$ ( )
   A) No change B) Increases C) Decreases D) Can't say
8. $\frac{2\tan 30°}{1+\tan^2 30°} =$ ( )
   A) $\sin 60°$ B) $\cos 60°$ C) $\tan 60°$ D) $\sin 30°$

9. $\dfrac{1+\tan^2 A}{1+\cot^2 A} =$ ( )

A) $\sec^2 A$ B) -1

C) $\cot^2 A$ D) $\tan^2 A$

10. If $\cot\theta$ is undefined then θ = ...

11. If $\sin\theta = \dfrac{a}{b}$ then $\tan\theta =$

A) $\dfrac{b}{a}$ B) $\dfrac{a}{\sqrt{b^2-a^2}}$

C) $\dfrac{a^2}{b^2}$ D) $\dfrac{b}{\sqrt{b^2-a^2}}$

12. If $x = \sin\theta + \cos\theta$, $y = \sin\theta.\cos\theta$ which of the following is true ?

A) $xy = 1$ B) $x^2 = 1 + 2y$

C) $xy = -1$ D) $x^2 = 1 - 2y$

13. If θ is acute angle then $\sin\theta \times \cos\theta \times \tan\theta \times \cot\theta \times \sec\theta \times \text{cosec}\,\theta$ = ...

14. $\cos\theta \times \sqrt{1+\tan^2\theta}$ = ...

15. ABC is an Isosceles right angled triangle $\angle B = 90°$, then $\tan^2 A - \cot^2 C$ = ...

9. SOME APPLICATIONS OF TRIGONOMETRY

[11 Marks] [1 + 2 + 8 = 11 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. From the top of a hill, a person observed the top and bottom of a 40m tall building with the angles of depression 30° and 45° respectively. Find the height of the hill and the distance between hill and the building.
2. The angle of elevation of the top of a tower from a point on the ground is 30° On moving 20m nearer to the tower, the angle of elevation increases to 45°. Find the height of the tower and the distance of the first point from the tower.'

2 MARKS QUESTIONS

1. Define the angle of elevation with a simple rough diagram.
2. The angle of elevation of the top of a tower from a point on the ground, which is 30m away from the foot of the tower is 30°. Find the height of the tower.
3. A tower standing vertically on the ground. From a point on the ground, which is 15m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° Find the height of the tower.
4. Name any two real life situations where trigonometry used.

1 MARK QUESTIONS

1. Define angle of Depression.
2. Assertion (A): when the Sun's altitude is 60°, The ratio of the length of shadow of a vertical tower to its height is $1:\sqrt{3}$.

   Reason (R) : tan 60° = $\sqrt{3}$

   A) Both Assertion(A) and Reason (R) are true and the reason (R) is the correct explanation of assertion (A)

   B) Both Assertion (A) and Reason (R) are true but the reason R is not the correct explanation of assertion (A)

   C) Assertion (A) is true and Reason (R) is false.

   D) Assertion (A) is false but Reason (R) is true.
3. Outline the diagram to the below situation :

   "A ladder of 10 m length touches a wall at a height 5m."

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. A building and a tower are standing on the same level ground. From the top of the tower, the angle of depression of the top and bottom of the building are 30° and 45° respectively. If the height of the tower is 80m, find:

   (a) The height of the building.

   (b) The horizontal distance between the tower and the building.

2. The angles of elevation of the top a tower from two points A and B on the same straight line from the base of the tower are complementary. If the distance between A and B is 50m, find the height of the tower and the distance of A and B from the base.

3. From a point on the ground the angles of elevation of the bottom and top of a transmission tower fixed at the top of a 40 m high building are 45° and 60° respectively. Calculate the height of the tower.

2 MARKS QUESTIONS

1. From the top of 7m high building, the angle of elevation of the top of the cable tower is 60º and the angle of depression of its foot is 45º. Determine the height of the tower.

2. The observer 1.5m tall is 28.5m away from a tower. The angle of elevation of the top of the tower from his eyes is 45°. What is the height of the tower?

1 MARK QUESTIONS

1. In heights and distance problems, angles are always taken with respect to line of sight. (True/False)

2. The angle of elevation of top of a pole of height 30m from a point $30\sqrt{3}$ m is 60°

   (True/False)

3. Koresh observed a flower pot on the ground from the balcony of the first floor of a building at an angle of depression θ, the height of the first floor of the building is p meters. Draw a rough diagram to this situation.

4. Define line of Sight.

10. CIRCLES

[7 Marks] [1 + 2 + 4 = 7 M]

LEVEL-1 : RISING STAR

4 MARKS QUESTIONS

1. Prove that the length of the tangents drawn from an external point to a circle are equal.
2. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that AB + CD = AD + BC.
3. Prove that the parallelogram circumscribing a circle is a rhombus.
4. Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

2 MARKS QUESTIONS

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre 'O' at a point Q so that OQ = 12 cm. Find the length of PQ.
2. From a point Q, the length of the tangent to circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.
3. The length of the tangent from a pont A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.
4. If TP and TQ are the two tangents to a circle with centre 'O' so that $\angle POQ = 110°$, then find $\angle PTQ$.

LEVEL-2 : SHINING STAR

4 MARKS QUESTIONS

1. Two tangents TP and TQ are drawn to a circle with centre 'O' from an external point T. Prove that $\angle PTQ = 2\ \angle OPQ$
2. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.
3. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle with touches the smallerl circle.

2 MARKS QUESTIONS

1. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of a contact.
2. Define tangent of a circle and secent of a circle.
3. A parallelogram ABCD is drawn to circumscribe a circle then write the relation between its sides.

1 MARK QUESTIONS

1. Create a geometrical design involving two tangents drawn from an external point to the circle.
2. Draw a circle and two lines parallel to a given line such that one is a tangent and the other a secant to the circle.
3. Genertate a geometrical design that the tangents drawn at the ends of a diameter of a circle are parallel.
4. Draw a rough sketch as a tangent of a circle passes through the end point of its radius.
5. Draw a rough sketch of a parallelogram circumscribing a circle.
6. Draw a circle inscribng in a quadrilateral.
7. How many tangents can a circle have ?
8. A circle can have ..... parallel tangents at the most. ( )

   A) 1 B) 2 C) 0 D) Infinite
9. The common point of a tangent to a circle and the circle is called .....
10. A tangent to a circle intersects it the circle ..... points.

13. STATISTICS

[12 Marks] [4 + 8 = 12 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. The following table gives the literacy rate (in percentage) of 35 cities. Find the mean literacy rate.

Literacy rate (in %)	45-55	55-65	65-75	75-85	85-95
No. of cities	3	10	11	8	3

2. 30 women were examined in a hospital by a doctor and the number of heart beats per minute were recorded and summarised as follows. Find the mean heartbeats per minute for these women, choosing a suitable method

No. of heart beats/minute	65-68	68-71	71-74	74-77	77-80	80-83	83-86
No. of workers	2	4	3	8	7	4	2

3. Consider the following distribution of daily wages of 50 workers of a factory.

Daily wages (₹)	500-520	520-540	540-560	560-580	580-600
No of workers	12	14	8	6	10

Find the mean daily wages of the workers of the factory by using an appropriate method.

4. The table below shows the daily expenditure on food of 25 households in a locality..

Daily expenditure	100-150	150-200	200-250	250-300	300-350
No.of households	4	5	12	2	2

Find the mean daily expenditure on food by a suitable method.

5. The distribution below gives the weights of 30 students of a class. Find the median weight of the students.

Weight (in kg)	40-45	45-50	50-55	55-60	60-65	65-70	70-75
Number of students	2	3	8	6	6	3	2

6. The following data gives the information on the observed lifetimes (in hours) of 225 electrical components.

Life times (in hrs)	0-20	20-40	40-60	60-80	80-100	100-120
Frequency	10	35	52	61	38	29

Determine the median lifetimes of the components.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarised it in the table given below. Find the mode of the data:

Number of cars	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	7	14	13	12	20	11	15	8

2. The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the median, mean and mode of the data

Monthly consumption (in units)	65-85	85-105	105-125	125-145	145-165	165-185	185-205
Number of consumers	4	5	13	20	14	8	4

3. The following distribution shows the daily pocket allowance of children of a locality. The mean pocket allowance is Rs.18. Find the missing frequency f.

Daily protect allowance (in ₹)	11-13	13-15	15-17	17-19	19-21	21-23	23-25
Number of childrens	7	6	9	13	f	5	4

4. If the median of the distrubtion given below is 28.5. Find the value of x and y.

C.I	0-10	10-20	20-30	30-40	40-50	50-60	Total
frequency	5	x	20	15	y	5	60

4 MARKS QUESTIONS

1. Write the formula for mode for grouped data? Explain each term in it.?
2. Write the formula for median for grouped data? Explain each term in it?
3. Write the formula for mean by assumed mean method? Explain each term in it?
4. Write the formula for mean by step deviation method? Explain each term in it?

14. PROBABILITY

[13 Marks] [8 + 4 + 1 = 13 M]

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting,
   i) a King of red colour ii) a face card iii) a Black face card.
   iv) the Jack of Diamonds v) a Spade vi) The Queen of Hearts
   vii) an ace of Red colour viii) not a Black of Jack
2. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 see figure below. And these are equally likely outcomes. What is the probability that it will point at

   i) '7' ? ii) An Even Number
   iii) A Number greater than '3' ? iv) A Number less than 9 ?
3. A box contains 90 discs which are numbered from 1 to 90.
   If one disc is drawn at random from the box, find the probability that it bears.
   i) a two-digit number ii) a perfect square number
   iii) a number divisible by '5'. iv) a perfect cube number
4. A die is thrown once. Find the probability of getting
   i) A prime number ii) a number greater than 4
   iii) An odd number iv) Factors of '2'

4 MARKS QUESTIONS

1. When a die is rolled once, find the probabilities of
   i) getting a number between 3 and 6, now create 4 such type of questions.
2. One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the drawn card will be "the king" ? Now create 4 such type of questions.
3. A box contains 5 red marbles, 8 white marbles and 4 green marbles. One marble is taken out of the box at random. What is the probability that the marble taken out will be green ? Now create 4 such type of questions.

1 MARK QUESTIONS

1. $P(E) + P(\overline{E}) =$ ..... ( )
   A) 0 B) – 1 C) 1 D) 1.5

LEVEL-2 : SHINING STAR

2. P(E) = 0.05 what is the probability of "not E" ?
3. Which of the following cannot be the probability of an event ? ( )
   A) $\frac{2}{3}$ B) 1.5 C) 15% D) 0.7
4. One card is drawn from a well shuffled deck of 52 cards. Calculate the probability that the card will (i) be an ace (ii) not be an ace.
5. The sum of the probabilities of all the elementary events of an experiment is .....( )
   A) 0 B) 1 C) 2 D) infinite
6. The probability of getting at least one Tail, when two coins are tossed is ..... ( )
   A) $\frac{3}{4}$ B) $\frac{1}{2}$ C) $\frac{2}{3}$ D) $\frac{1}{4}$
7. If a letter is taken at random from the letters in the word "MOBILE", what is the probability that it will be a "consonants" letter ? ( )
   A) 1 B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{6}$
8. The probability of getting a number less than 4 on the top when rolling a die is ( )
   A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{5}{6}$ D) 0.75
9. A card is drawn at random from a well-shuffled deck of 52 cards. Calculate probability of a Red King ? ( )
   A) $\frac{1}{13}$ B) $\frac{2}{13}$ C) $\frac{1}{26}$ D) $\frac{1}{52}$
10. A bag contain 3 green marbles, 4 blue marbles and 2 orange marbles. What is the probability that a coloured marble drawn at random will not be an green marbles. ( )
    A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{4}{9}$ D) $\frac{7}{9}$
11. What is the probability that a number is chosen at random from a two-digit number and that it is a multiple of 3 ? ( )
    A) $\frac{1}{3}$ B) $\frac{3}{10}$ C) $\frac{7}{25}$ D) $\frac{29}{100}$
12. The probability of getting more than 5 on the top of a die is ( )
    A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{1}{6}$ D) $\frac{3}{5}$
13. The probability that a die rolled will result in a prime number is ( )
    A) $\frac{2}{3}$ B) $\frac{1}{2}$ C) 1 D) $\frac{1}{6}$

14. If a number is drawn at random from the 8 numbers 1, 2, 3, 4, 5, 6, 7, 8 what is the probability that it is a factors of '6' ? ( )

A) $\frac{1}{2}$ B) $\frac{5}{6}$ C) $\frac{3}{4}$ D) $\frac{7}{8}$

15. If 3 coins are tossed at the same time, the probability of not getting heads is ( )

A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) $\frac{7}{8}$ D) $\frac{1}{8}$

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Two dice are thrown at the same time.

   i) Write all possible outcomes.

   ii) What is the probability that sum of two numbers appearing on the top of the dice

   a) 7 b) 16

   iii) Find the probability of same number on both sides.

2. A game consists of tossing a one rupee coin '3' times and nothing its outcome each time.Hanif wins if all the tosses give the same result i.e., three heads or three tails, and losses otherwise. Calculate the probability that Hanif will lose the game.

3. A bag contains cards with numbers from 1 to 36 written on them. After mixing the cards well, a card is choosen at random from them and the probability that it is the following card is calculated.

   i) An odd number ii) Prime number iii) Perfect squares

   iv)Multiples of 3 v) Only Even composite number

   vi) The number is neither prime number nor a composite number.

   vii) Factors of 24 viii) Only an odd prime number

4. Five cards. the ten, Jack, queen, king and ace of diamonds, are well-suffled with their face downwards. One card is then picked up at random.

   i) What is the probability that the card is the queen

   ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?

4 MARKS QUESTIONS

1. Two dice are thrown at the same time. What is the probability that the sum of two numbers appearing on the top of the dice is '7' ? Now create 4 such type of questions.

2. A coin tossed 4 times. Find the probability of getting all the tails ? Now create '4' such type of questions.

3. A bag contains 3 yellow balls, 6 green balls 4 red balls and 2 white balls. One ball is taken out of the bag at random. What is the probability that the ball take out will be a red ball ? Now create 4 such type of questions.